CHAPTER ONE

MEASUREMENT OF PHYSICAL QUANTITIES

1.0 Introduction

Physics is the branch of science which deals with the nature and properties of matter and its relation to energy. It involves the study of matter based on quantitative analysis using mathematics as a tool to finding physical laws that govern the behavior, characteristics and the nature of everything from the quantum (microstate) particles of the nano world to the planets, solar systems and galaxies that occupy the cosmos.

Prospects in Physics

The knowledge of physics is broadly utilized in the society of today ranging from industries, institutions, meteorology, to health sector. Career directly related to physics degree include: Meteorologist, Nanotechnologist, Radiation protection practitioner, Scientific researcher, Educationalist, Sound engineer, Technical aut Metallurghor, .Acoustic consultant, Astronomer, Clinical scientist, Medical physicist, Geophysicist, Academician etc. Physics degrees are applicable in various establishments such as:Federal Radiation Protection and Safety (FRPS), National Meteorological Agency (NIMET), Center for Raw Material Research, Center for Solid Minerals Research.Federal geological survey, NNPC and other Nigeria Oil Companies. Medical / Radiation physicist, Pollution Specialist/ Environmental physicists.

Telecommunication Industries, Software and Computing, consultancy industries and Lecturer or Professional Teacher.

1.1 Measurements

Measurements play an important role in Physics, it gives a numerical value to the property of an item. As an illustration, it is simple to determine who is taller when two people are placed side by side. However, it would be hard to compare the heights of the two persons if they are not physically present in the same location. The basic analogy discussed above was used to carry out measurement in the olden days. But it would be quiet challenging to obtain accurate result if the objects under measurement are not together for comparison . Measurement of Physical quantities are represented with numerical Values and Units

A complete set of units for all kinds of physical quantities are called system of units (fundamental and derived units). Uniformity of the results in science is inevitable and it can be achieved only through the measurement standardization. The measurement is expected to be reliable, although no

measurement can be precise due to uncertainty associated with every re
This emanated from the limitation of measuring instrument (systematic erro
and the inability of personnel to carry out measurement in accordance wi
the lay-down rule and regulation governing the operation (random error).

1.2 Physical Quantity

Physical quantity is the measurable property of a material. Measurement
necessary to determine magnitude of a physical quantity to compare
similar physical quantities and to prove physical laws or equations. A physi
quantity is represented completely by its magnitude and unit.

Physical quantity can be expressed as the product of magnitude and unit .i.e

For instance, 10 *meter (m) means* a length which is ten times the unit lengt
m. Here 10 represent the numerical value of the given quantity and *me*
represents the unit of quantity under consideration. A physical quantity c
be in either fundamental or derived form.

1.3 Fundamental and Derived Quantities

A. *Fundamental quantities*

Physical quantities such as Mass, Length, Time, etc. which exist in nature a
are independent on the other quantities are called fundamental or ba
quantities. The following are the seven (7) basic quantities which a
independent of other quantities and can be expressed in terms of their symbo

S/N	*Quantity*	*Unit*	*Abbreviation*
1	Length	Meter	m
2	Mass	Kilogram	kg
3	Time	Seconds	s
4	Electric current	Ampere	A
5	Temperature	Kelvin	K
6	Amount of substance	Mole	mol
7	Luminous intensity	Candela	cd

B. *Derived quantities*

Derived quantities belong to the categories of physical quantities whi
depends on the seven basic quantities enlisted above. They are derived
multiplication or division of different fundamental quantities. For instanc
if length is defined as a fundamental quantity, then area and volume a
derived from length and are expressed in terms of length.

It should be noted that in mechanics *Length, Mass and Time* are arbitrarily chosen as fundamental quantities. However, these set of fundamental quantities are not a unique choice. In fact, any three quantities in mechanics can be termed as fundamental as all other quantities in mechanics can be expressed in terms of these. *e.g.* if speed and time are taken as fundamental quantities, length will become a derived quantity because then length will be expressed as Speed x Time. There are several derived quantities

S/N	Physical Quantity	Formula	Unit
1.	Velocity (v)	$\frac{\text{Displacement}}{\text{Time}}$	m/s
2.	Acceleration	$\frac{\text{Velocity}}{\text{time}}$	m/s/s ms^{-2}
3.	Force (F)	mass × acceleration	$kgms^{-2}$
4.	Impulse	force × time	$kgms^{-1}$
5.	Momentum (P)	mass × vel.	$kgms^{-1}$
6.	Work	force × dist.	$kgms^{2}s^{-2}$
7.	Density	$\frac{\text{mass}}{\text{Volume}}$	kgm^{-3}
11.	Coefficient of viscosity	$\frac{\text{Tensile stress}}{\text{Velocity gradient}}$	$kgm^{-1}s^{-1}$
12.	Electric potential	$v = \frac{\Delta W}{q}$	$kgm^{2}s^{-3}A^{-1}$
13.	Capacitance	$Q = cv$	$kg^{-1}m^{-2}T^{4}A^{2}$
14.	Resistance	$V = IR$	$kgm^{2}s^{-3}A^{-2}$
15.	Conductance	$S = \frac{1}{R}$	$kg\text{-}1m^{-2}S^{-3}A^{2}$

It should be noted that when a physical quantity is derived as a ratio of two similar quantities, it gives no unit (i.e. it is unitless). The common *examples of these quantities are:*

1. *Relative desity* $= \frac{\text{Density of object}}{\text{Density of water at 4 °C}} = 1$
2. *Refractive index* $= \frac{\text{Velocity of light in air}}{\text{Velocity of light in a medium}} = 1$
3. *Strain* $= \frac{\text{Change in length}}{\text{Original length}} = 1$

1.4 Dimensional Analysis

The study of the relationship between physical quantities based on their and dimensions is known as *dimensional analysis*. It is used to change form of a unit into another. *Units* are very useful when trying to an equations, complicated formulas can be analyzed by checking its consis simply by considering the units (dimensions) of all the quantities to e that both sides of the equation equal. It is important to remember that th symbol has a very specific meaning in mathematics and physics. It means whatever is on either side of this sign is exactly the same thing even tho may look a little different on either side. If both sides must be the same they must also have the same *units*. The basic strategy is to represe quantities in the equation by their dimensions, but before the dimensio different basic quantities are to be written one need

to establish dimension of three basic quantities *Length*(L), *Time*(T), *Mass*(M) as foundational way of determining the dimension of e physical quantity or uniqueness of the equation.

The following are the dimensions of some important physical quantities

S/N	Physical Quantity	Unit	Dimension
1.	Velocity	ms^{-1}	LT^{-1}
1.	Acceleration	ms^{-2}	LT^{-2}
2.	Force (F)	$kgms^{-2}$ or N	MLT^{-2}
3.	Impulse	Nm or $kgms^{-1}$	MLT^{-1}
4.	Momentum (P)	$kgms^{-1}$	MLT^{-1}
5.	Work	kgm^2s^{-2}	ML^2T^{-2}
6.	Density	kgm^{-3}	ML^{-3}
7.	Coefficient of viscosity	$kgm^{-1}s^{-1}$	$ML^{-1}T^{-1}$

1.5 Application of Dimensional Analysis

1. ***To find the unit of a physical quantity in a given system of units***

The dimension of physical quantity can be determined through the definit or formula to calculate the quantity.

Example 1.1

The dimension of work can be obtained by writing its formula and unit as shown below:

Work (w) = force × distance

Work (w) = $[kgms^{-2}] \times [m]$

Work (w) = $[ML1T^{-2}] \times [L1]$

Work (w) = $[ML^2T^{-2}]$

Dimensionally, work is ML^2T^{-2}

2. ***To find dimensions of physical constant or coefficient***

The dimensions of a given constant incorporating in the formula due to proportionality sign can be determine through the formula as shown below.

Example 1.2

The dimension of gravitational constant (G) according to Newton's law of gravitation:

$$F = G\frac{m_1m_2}{r^2}$$

$$G = \frac{Fr^2}{m_1m_2}$$

$$G = \frac{N.m^2}{Kg.kg}$$

$$G = \frac{Kg.\frac{m}{s^2}m^2}{Kg.kg}$$

$$G = \frac{Kg.m^3s^{-2}}{Kg.kg}$$

$$G = Kg^{-1}m^3s^{-2}$$

Dimensionally, $G = M^{-1}L^3T^{-2}$

3. ***To check the dimensional correctness of a given physical relation***

This application is otherwise known as the '*principle of homogeneity*'. According to this principle the dimensions and terms on both sides of an equation must be the same.

Example 1.3

i. If $F = \frac{Mv^2}{r}$, by substituting dimension of the physical quantities in the above relation.

$$[M\,L\,T^{-2}] = [\frac{M\,L^2\,T^{-2}}{L^2}$$

$$[M\,L\,T^{-2}] = [M\,T^{-2}]$$

This implies that the formula is dimensionally correct.

Example 1.4

ii. Check whether the equation below is dimensionally correct.

$$S = ut + \tfrac{1}{2}at^2$$

Substitute the dimension of the physical quantities in the above relation.

$$[L] = [LT^{-1}][T] + \tfrac{1}{2}\,[LT^{-2}][T^2]$$

$$[L] = [L] + \tfrac{1}{2}\,[L]$$

$$[L] = \tfrac{3}{2}\,[L]$$

This equation is dimensionally correct. $\frac{3}{2}$ is a dimensional constant.

4. ***As a research tool to derive new relations***

If the relationship between the quantities are known, then one can be derive its govern formula.

Example 1.5

The time (t) of a simple pendulum is a function of mass of the bob (m), effective length (l) and acceleration due to gravity (g). Find the expression time T.

Assume that the function is a product of power function of m, l and g.

$$T = km^x l^y g^z$$

Substituting the dimensions of quantities. then

$$[T] = [M^x][L]^y\,[LT^{-2}]^z$$

Where k = dimensionless constant

$$M^0L^0T^1 = M^XL^{Y+Z}T^{-2Z}$$

by equating the exponents of like quantities

$$X = 0,\ Y + Z = 0 \text{ and } -2Z = 1$$

solving for x, y and z, we have $x = 0, y = 1/2$ and $z = -\frac{1}{2}$. Substitute values of x, y and z in

$$T = km^{x}l^{y}g^{z}$$

$$(T) = k.M^{0}.l^{1/2}g^{-1/2}$$

$$T = kl^{1/2}g^{-1/2}$$

$$T = \frac{kl^{1/2}}{g^{1/2}}$$

$$T = k\sqrt{\frac{l}{g}}$$

So the required physical relation becomes. $T = 2\pi\sqrt{\frac{l}{g}}$ the value of

Dimensionless constant is found (2π) through experiment.

Example 1.6

When a small sphere moves at low speed through a fluid, the viscous force ***F***, opposing the motion, is found experimentally to depend on radius (***r***), velocity (v) of the sphere and viscosity of the fluid.

Solution

From the question $F \alpha k \eta r v$

Introduce x, y and z as powers of

$$F = k\eta^{x}r^{y}V^{z}$$

Where *k* is a dimensionless constant, by using dimensional relation, we have.

$$M^1L^1T^{-2}] = K[M^1L^{-1}T^{-1}]^x[L^1]^y[L^1T^{-1}]^z$$

$$M^1L^1T^{-2}] = kM^xL^{-x+Y+Z}T^{-X-Z}$$

by solving for , we have the following equations,

$$1 = x,\ 1 = -x + y + z \text{ and } -2 = -x - z$$

Where $x = y = z = 1$, equation become

$F = k\eta rv$

Experimentally, $k = 6\pi$, so, $F = 6\pi\eta rv$ (Stoke's Law)

Example 1.7

If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units and are directly proportional to the mass (m), find the dimensions of mass in new system.

Solution

Let $M \propto C^x G^y h^z$ then.

$$M = k\, C^x G^y h^z \ \ldots\ldots \text{eqn. } (*)$$

where k is the constant of proportionality

$$M^1L^0T^0 = k[LT^{-1}]^X.[M^{-1}L^3T^{-2}]^Y.[M^1L^2T^{-1}]^z$$

$$M^1\ L^0\ T^0 = kM^{-y+z}\ L^{x+3y+2z}\ T^{-x-2y-z}$$

equating the powers on both sides, we have

$$1 = -y + z,\ 0 = x + 3y + 2z \text{ and } 0 = -x - 2y - z.$$

Solving for x, y and z in the above equation we obtain $x = \frac{1}{2}$, $y = \frac{1}{2}$ and $Z = 1/2$ Hence, equation (*) becomes

$$M = kc^xG^yh^z = kc^{\frac{1}{2}}G^{\frac{1}{2}}h^{\frac{1}{2}}, \text{ then } M = k\sqrt{cgh}$$

Example 1.8

If the period (T) of vibration of a liquid drop depends on surface tension (s), radius (r) of the drop and density (e) of the liquid, find the expression for T.

Solution

Let $T \propto s.r.p.$

$$T = ks.r.p.$$

Where k is the constant of proportionality

Let $T = ks^xr^yp^z$

substituting the dimension of each quantity in both sides, we have

$$M^0L^0T^1 = k[M^1T^{-2}]^x[L^1]^y[M^1L^{-3}]^z$$

$$M^0L^0T^1 = kM^{x+z}L^{y-3z}T^{-2x}$$

equating the exponents of both sides, we have

$$0 = x + z, \quad 0 = y - 3z \text{ and } 1 = -2x$$

Hence by solving for x, y and z, we have

$x = -1/2, y = 3/2$ and $z = 1/2$ Hence equation becomes

l ks r

$$T = k\sqrt{\frac{p\,r^3}{s}}$$

COMMON S.I. PREFIX MULTIPLIERS

Prefix	Symbol	Multipliers	Power of 10
Giger	G	1, 000. 000,000	10^9
Mega	M	1, 000, 000	10^6
Kilo	k	1, 000	10^3
Deci	d	0.1	10^{-1}
Centi	c	0.01	10^{-2}
Milli		0.001	10^{-3}
Micro	μ	0.000001	10^{-6}
Nano	n	0.000000001	10^{-9}

Example 1.9

Use the prefix multipliers above to express each of the following measurement into basic form.

i. 100 μm
ii. 500 μF
iii. 0.001Gb

Solution

i. 100μm 100 $10^{-6}=1\times10^{-4m}$

ii. 500 μF = *500* $\times10^{-6}$ F = 5×10^{-4}m

iii. 0.001Gb = $0.001\times10^{9}=10^{6}$b

Exercise 1

1. Derive the unit of the following quantities:

A. Resistivity [Ω *ans*: m or $kgm^3s^{-3}A^{-2}$]
B. Conductivity [$\Omega^{-1}m^{-1}$ or $kg^{-1}.m^{-3}s^3A^2$] c
C. permittivity of free space 'ε; [*ans*: NC^2m^{-2} or $kgm^{-3}s^{-2}C^2$]

2. Find the dimension of the following physical quantities:
   a. Stress [*ans*: $ML^{-1}T^{-2}$]
   b. Young modulus [*ans*: $ML^{-1}T^{-2}$]
   c. Surface tension [*ans*: MT^{-2}]
   d. Momentum [*ans*: MLT^{-1}]

3. Derive the dimension of the following physical quantities:
   a. Energy [*ans*: ML^2T^{-2}]
   b. Moment. [*ans*: ML^2T^{-2}]
   c. Frequency [*ans*: T^{-1}]
   d. Power [*ans*: ML^2T^{-3}]
   e. Pressure [*ans*: $ML^{-1}T^{-2}$]

4. The volume *V* of water passing through a point of a uniform tube during *t* seconds is related to the cross sectional area *A* of the tube and velocity *u* of water by the relation $V \alpha A^{\alpha} u^{\beta} t^{\gamma}$, which one of the following will be true. [*Ans*: $\alpha \neq \beta = \gamma$]

5. Kinetic energy *K* has dimensions $kg.m^2/s^2$ It can be written in terms of the momentum *p* and mass *m* as $k = \frac{p2}{2m}$

   a. Determine the proper units for momentum using dimensional analysis.[$kg^{1/2}m^{1/2}s^{-1}$]
   b. The unit of force is the newton N, where $1N = 1kg.m/s^2$?. What are the units of momentum *p* in termsof a newton? [$N^{1/2}m$]

6. Express dimension of force and charge in terms of the S.I base unit kilogram, meter, seconds and ampere. Hence, with reference to coulombs law

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$$

7. A certain physical law is given by where T is the period in seconds, k is dimensionless, d is the density, r is the distance and S is the Coefficient of surface tension. Determine a, b and c [*Ans*: $a = \frac{1}{2}$ $b = \frac{3}{2}$ $c = -\frac{1}{2}$]

8. Newton's law of universal gravitation is represented by. $F = \frac{G M m}{r^2}$ where *F* is the magnitude of thegravitational force exerted by one small object on another, *M* and *m* are the masses of the objects, and *r* is a distance. Force has the SI units $kg.m/s^2$?. What are the SI units and dimension of the proportionality constant G [*Ans:* $kg^{-1}m^3s^{-2}$ and $M^{-1}L^3T^{-2}$.]

9. Use dimensional analysis to determine the value of 'L' in the relation, $T = K a^k p^L \gamma^m$ T is the period of vibration, a is the radius, p is the density And is the surface tension.[*Ans:* $L = \frac{1}{2}$]

10. show that the expression $v^2 = u^2 + 2as$ is dimensionally correct, where v and u represent the final and initial velocities, a is the acceleration and s is the displacement. [*ans:* Dimensionally correct].

11. The period T of a loaded spring-oscillator depends on its mass m and spring constant k, as follow $T = C K^x m^y$ where x and y are integers and C is a dimensionless constant. Find the value of x and y. [*Ans:* $x = -\frac{1}{2}$ and $y = \frac{1}{2}$]

12. The experiment shows that the frictional drag force F, of a car moving in air be influenced by the velocity v of its motion, density p of the air through which it travel, and the cross-sectional area of the car, if the force $F = C P^k v^L A^m$ where k, L and m are the integers and C is a dimensionless constant, find the value of k, L, and m. [*Ans:* $K = 1, L = 1 \text{ and } m = 1$]

CHAPTER

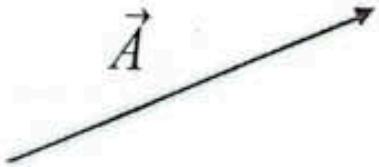
VECTOR AND SCALAR QUANTITIES

2.0 Introduction

Apart from the Fundamental and derived quantities classification, physical quantities can also be classified into another supplementary group called scalar and vector quantities. Scalar quantiles are the quantities which can be described in terms of magnitude only **(i.e. *without Direction*)** such as distance, speed, power, energy, temperature, etc. Vector quantities are the quantities which has both magnitude and direction like displacement, velocity, and acceleration, etc. The motion of a bicycle along a straight line can be determine by its speed covered (scalar) and its direction of its motion (vector). Usually its very important in mechanics to describe and represent a vector by its magnitude and direction for a complete information about the body.

2.1 Vector Representation

Generally, vector quantities can be represented on the paper with capital letter well bold(**A**) or presented in italic form ($\vec{A}$). It can also be written in capital letter with an arrow display on its head $|\mathbf{A}|$. Similarly, vector magnitude or absolute value can be written inform of *(A)* . Its graphical representation is as shown below with the length of the arrow representing the magnitude and the direction of the arrow giving the direction of the vector.

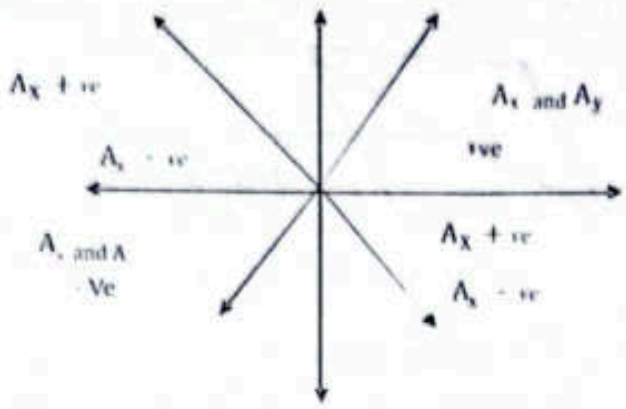


Vector representation

This graphical representation might help the basic arithmetic understanding of vectors. Mathematically, we denote a vector $\vec{\mathbf{A}}$, with an arrow over the variable. If only the variable use without an arrow, it implies that we are referring to the magnitude of the vector.

1.2 Vector Direction Sign Graph

The direction of the vector depends on the quadrant and location of both x and y axis at that instance. It is possible for the component of vector to be either positive or negative depending on the angle (quadrant) through which the vector positioned. Consider a positive vector *A in the figure below,produced the positive components along both x and y axis in the first quadrant. But the components only turned negative in the second quadrant when the angle is between 90 and 180 degrees.*



1.3 Vector Properties

Equality of vectors

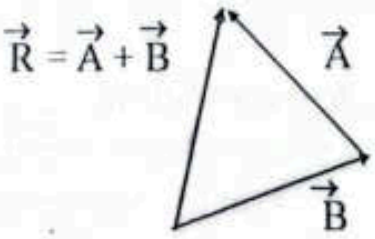
Two vectors are equal only if they have the same magnitude and direction (two arrows are the same only if they are the same length and point in the same direction). This means that you can move a vector around in space as long as you don't change the length or direction of the vector.

Addition of Vectors

Just as when you are adding scalar quantities, you must ensure that the vectors you are trying to add have the same units. Vectors can be added either geometrically (graphically) or algebraically.

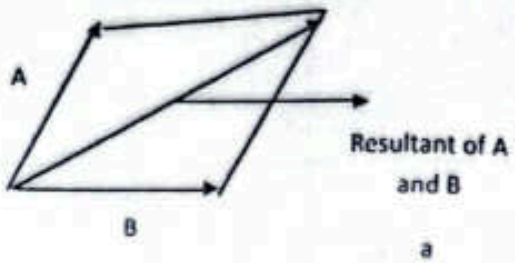
Graphically addition of two vectors $\vec{A}$ and $\vec{B}$.

The vectors can be added by draw the vectors (using the same scale for both) head to tail on a piece of paper. The resultant vector $\vec{R} = \vec{A} + \vec{B}$ is the vector drawn from the unmatched tail to the unmatched head. Graphical addition of vectors is useful for visualizing the addition process.



Algebraically addition of two vector

Resolve the vectors into their x- and y- components. All the x-components are added to get the x-component of the resultant vector. All the y-components are added to get the y component of the resultant vector. Never add the x-components to y-components. The magnitude and direction of the resultant vector can be obtained by converting from the Cartesian representation (x and y components) using the equations.



a

Negative of a vector

The negative of a vector, $\vec{A}$, is define as the vector that gives zero when add to positive vector $\vec{A}$. i.e. the negative of $\vec{A}$ has the same magnitude as $\vec{A}$, b opposite direction (difference).

Subtracting vectors

Subtraction is simply the addition of a negative quantity. Since we h defined the negative of a vector, we can figure out how to subtr Remember that the negative of a vector points in the opposite direction, so subtraction we flip the direction of the vector to be subtracted, but otherw use the same graphical method as for addition. Graphically flipping a vec corresponds to algebraically changing the sign of both the x- and components of the vector, so we can use the same algebraic method subtracting vectors as we do for adding vectors.

$$\vec{A} = \vec{R}, \vec{B},$$

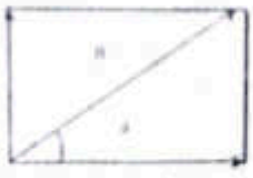
1.4 Parallelogram Law of Vector

The law states that if two vectors are drawn from a point as two adjacent sid of a parallelogram, the resultant is the diagonal of the parallelogram. Th pictorial vector below shown the parallelogram of vector rule.

Example 2.1

Consider vectors A and B acting at right angles to each other, a Pythago theorem can be used to find the resultant R.

Using Pythagoras Theorem Method



B

$$R = \sqrt{A^2 + B^2}$$

By taking A and B to be 30 and 40 respectively, we have

$$R = \sqrt{30^2 + 40^2}$$

$$R = \sqrt{2500}$$

$$R = 50\ N$$

The direction of the resultant with respect to a given line OB. It is given as;

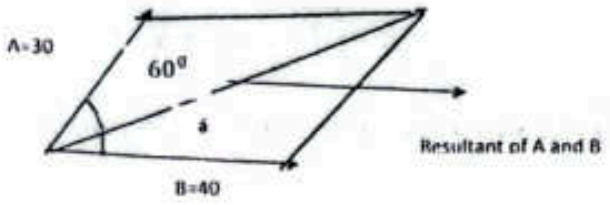
$\tan\alpha = \frac{\text{opposite}}{\text{Adjacent}}$ $\frac{A}{B}$ $\tan a = \frac{30}{40}$ $a = 37^0$ Note that the graphical Method can also be used to determine the resultant.

Example 2.2:

In the figure below, A and B are the vectors representing the forces 30N and 40N respectively. Completing the parallelogram above gives R as the resultant force, then resultant R can be determining as:

$R^2 = A^2 + B^2 + 2AB\cos 60^0$

If $A = 30$ and $B = 40$



$$R^2 = 30^2 + 40^2 + 2(30)(40)\cos 60^0$$

$$R = \sqrt{3700} = 60.8$$

Using the sine rule

$$\frac{\text{Sin}a}{A} = \frac{\text{Sin } 120^0}{R}$$

If $R = 60.8N$, $A = 30N$

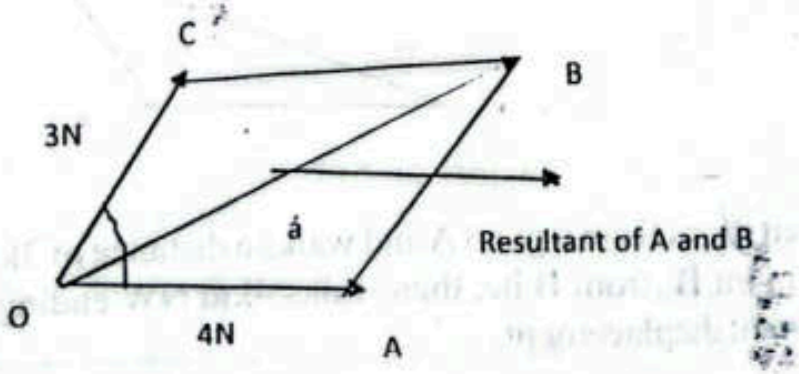
$$\frac{\text{Sin}a}{30} = \frac{\text{Sin } 120^0}{60.8}$$

$\sin\alpha = 0.4273\alpha = 25.3^0$ hence, the Resultant makes an angle of with the horizontal force 40N.

Example2.3

Two forces 3N and 4N acts on the body. If the angle between the force is.60° calculate the magnitude of the resultant on the body and its direction

Solution



In the figure above, OA and OC are the vectors representing the force 3N and 4N respectively. Completing the parallelogram OABC gives OB as the resultant force. Since OC is parallel to AB, ROAB = 180° - 160°

By applying cosine rule, we have

$$R^2 = OA^2 + Oc^2 + (OA)\ (OC)\cos 60^\circ$$

If $OA = 3$ *and* $OC = 4$

$$R^2 = 3^2 + 4^2 + 2\,(3)\,(4)\cos 60^\circ$$

$$R = \sqrt{37} = 6.08\,N$$

Applying sine rule to determine direction

$$\frac{\text{Sin}a}{OA} = \frac{\text{Sin } 120^\circ}{R}$$

But R = 6.08N, A = 3N

$$\frac{\text{Sin}a}{3} = \frac{\text{Sin } 120^\circ}{6.08}$$

$$\alpha = 25.3^0$$

2.5 The Triangle Law of Vector Addition

The triangle law of vector addition states that going from **A** to **B** then from B to **C** is the same as going directly from **A** to **C** as shown in the diagram below. Remember that the second vector begins at the end of the first vector when we use the triangle law. The cosine rule, sine rule or Pythagoras theorem (for right-angled triangle) may also be applied to the triangle.

Example 2.4



A man starts from a point A and walks a distance of 3km in a direction due NE to a point B, from B he, then walks 4km NW ending at a point C. Find his resultant displacement.